

## HALLEY STATION, Antarctica

WMO: 890220

Lat: 75.607S

Lon: 26.208W

Elev: 30

StdP: 100.97

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |              |             |              | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|--------------|-------------|--------------|-----------------------|-----------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |              | 1%          |              |                       |           |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB         | WS          | MCDB         | MCWS                  | PCWD      |              |
| (a)<br>7      | (b)<br>-45.5 | (c)<br>-43.1 | (d)<br>-49.3                  | (e)<br>0.0 | (f)<br>-45.1 | (g)<br>-47.0 | (h)<br>0.0 | (i)<br>-43.0 | (j)<br>23.5           | (k)<br>-18.9 | (l)<br>21.4 | (m)<br>-18.6 | (n)<br>2.8            | (o)<br>90 | (p)<br>1.358 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |           |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|-----------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |           |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD      |
| (a)<br>1      | (b)<br>5.8             | (c)<br>-0.1     | (d)<br>-0.9 | (e)<br>-0.8 | (f)<br>-1.5 | (g)<br>-1.5 | (h)<br>-2.1 | (i)<br>-0.7         | (j)<br>-0.4 | (k)<br>-1.3 | (l)<br>-0.9 | (m)<br>-2.0 | (n)<br>-1.6 | (o)<br>6.1           | (p)<br>70 |

| Dehumidification DP/MCDB and HR |     |      |      |     |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|------|-----|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%   |     |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP   | HR  | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d)  | (e) | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| -1.0                            | 3.5 | -0.7 | -1.8 | 3.2 | -1.2 | -2.7 | 3.0 | -1.8 | 8.5           | -0.3 | 7.4  | -0.9 | 6.2  | -1.5 | 3.5               |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |         |                    |          | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|----------|----------|----------------------------|---------|--------------------|----------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |          |          | Mean                       |         | Standard Deviation |          | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max     | Min                | Max      | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 19.9          | (b) 17.6 | (c) 14.9 | (e) -48.8                  | (f) 1.5 | (g) 2.8            | (h) 1.1  | (i) -50.8                                          | (j) 2.3  | (k) -52.5  | (l) 3.0  | (m) -54.0  | (n) 3.6  | (o) -56.0  | (p) 4.5  |    |    |
| (4) 19.9          | (5) 17.6 | (6) 14.9 | (7) -48.9                  | (8) 0.1 | (9) 2.8            | (10) 0.7 | (11) -50.8                                         | (12) 0.6 | (13) -52.5 | (14) 1.0 | (15) -54.0 | (16) 1.4 | (17) -56.0 | (18) 2.0 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | -19.5  | -5.3  | -11.2 | -17.7 | -23.6 | -25.3 | -27.5 | -29.0 | -28.9 | -26.9 | -21.1 | -11.7 | -5.9  |  |  |
|                                                                        | DBStd       | 10.62  | 2.63  | 4.39  | 6.35  | 7.52  | 7.28  | 8.33  | 7.43  | 7.63  | 7.56  | 6.00  | 4.23  | 2.67  |  |  |
|                                                                        | HDD10.0     | 10781  | 473   | 595   | 858   | 1008  | 1095  | 1126  | 1209  | 1206  | 1107  | 963   | 650   | 492   |  |  |
|                                                                        | HDD18.3     | 13822  | 731   | 828   | 1117  | 1258  | 1353  | 1376  | 1467  | 1464  | 1357  | 1221  | 900   | 750   |  |  |
|                                                                        | CDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 5.8    | 4.6   | 5.1   | 5.8   | 5.7   | 5.8   | 6.4   | 6.0   | 5.9   | 6.5   | 6.4   | 6.5   | 5.1   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 295    | 18    | 21    | 23    | 20    | 23    | 26    | 25    | 28    | 32    | 26    | 32    | 20    |  |  |
|                                                                        | PrecMax     | 461    | 37    | 56    | 51    | 41    | 58    | 73    | 54    | 53    | 78    | 91    | 109   | 58    |  |  |
|                                                                        | PrecMin     | 188    | 4     | 2     | 8     | 3     | 6     | 7     | 6     | 3     | 4     | 2     | 5     | 6     |  |  |
|                                                                        | PrecStd     | 66     | 9     | 11    | 9     | 10    | 14    | 19    | 13    | 12    | 19    | 17    | 21    | 13    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 1.0   | -0.8  | -2.2  | -4.9  | -4.1  | -7.1  | -11.1 | -7.3  | -8.0  | -6.3  | -1.7  | 0.6   |  |  |
|                                                                        |             | MCWB   | -0.1  | -1.2  | -2.3  | -5.2  | -4.2  | -7.6  | -11.3 | -7.4  | -8.5  | -6.6  | -2.4  | -0.7  |  |  |
|                                                                        | 2%          | DB     | 0.1   | -1.9  | -4.4  | -6.9  | -9.0  | -9.7  | -12.8 | -11.2 | -10.5 | -8.0  | -3.2  | -0.3  |  |  |
|                                                                        |             | MCWB   | -0.6  | -2.3  | -4.7  | -7.2  | -9.3  | -10.0 | -13.1 | -11.6 | -10.8 | -8.3  | -3.8  | -1.1  |  |  |
|                                                                        | 5%          | DB     | -0.6  | -3.1  | -6.1  | -8.8  | -12.3 | -12.2 | -15.3 | -14.5 | -12.5 | -9.6  | -4.2  | -0.9  |  |  |
|                                                                        |             | MCWB   | -1.2  | -3.5  | -6.4  | -9.1  | -12.5 | -12.4 | -15.5 | -14.8 | -12.8 | -9.9  | -4.7  | -1.7  |  |  |
|                                                                        | 10%         | DB     | -1.2  | -4.4  | -8.0  | -11.1 | -14.7 | -14.9 | -17.9 | -17.3 | -15.2 | -11.5 | -5.3  | -1.7  |  |  |
|                                                                        |             | MCWB   | -1.8  | -4.8  | -8.4  | -11.4 | -14.9 | -15.1 | -18.1 | -17.5 | -15.4 | -11.8 | -5.7  | -2.4  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 0.2   | -0.9  | -2.2  | -5.1  | -4.1  | -7.4  | -11.3 | -7.4  | -8.3  | -6.5  | -2.3  | -0.1  |  |  |
|                                                                        |             | MCDB   | 0.5   | -0.8  | -2.2  | -4.9  | -4.1  | -7.2  | -11.2 | -7.2  | -8.2  | -6.4  | -1.9  | 0.1   |  |  |
|                                                                        | 2%          | WB     | -0.4  | -2.2  | -4.5  | -7.1  | -9.3  | -9.9  | -13.1 | -11.4 | -10.8 | -8.3  | -3.6  | -0.9  |  |  |
|                                                                        |             | MCDB   | -0.1  | -2.0  | -4.4  | -6.9  | -9.0  | -9.6  | -12.9 | -11.1 | -10.6 | -8.1  | -3.3  | -0.5  |  |  |
|                                                                        | 5%          | WB     | -1.0  | -3.4  | -6.4  | -9.1  | -12.6 | -12.4 | -15.5 | -14.8 | -12.7 | -9.8  | -4.7  | -1.6  |  |  |
|                                                                        |             | MCDB   | -0.7  | -3.1  | -6.1  | -8.8  | -12.4 | -12.2 | -15.3 | -14.5 | -12.4 | -9.5  | -4.4  | -1.1  |  |  |
|                                                                        | 10%         | WB     | -1.7  | -4.8  | -8.4  | -11.4 | -14.9 | -15.1 | -18.0 | -17.5 | -15.4 | -11.8 | -5.7  | -2.3  |  |  |
|                                                                        |             | MCDB   | -1.2  | -4.5  | -8.1  | -11.1 | -14.7 | -14.9 | -17.9 | -17.3 | -15.2 | -11.5 | -5.3  | -1.8  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 5.8   | 7.5   | 8.1   | 8.6   | 8.5   | 8.4   | 8.4   | 8.6   | 8.4   | 8.8   | 7.3   | 6.0   |  |  |
|                                                                        |             | MCDBR  | 5.8   | 6.4   | 6.8   | 7.4   | 8.5   | 8.3   | 8.8   | 8.6   | 7.1   | 8.1   | 6.7   | 6.1   |  |  |
|                                                                        |             | MCWBR  | 5.2   | 6.2   | 6.8   | 7.5   | 8.5   | 8.3   | 8.8   | 8.6   | 7.1   | 8.0   | 6.3   | 5.6   |  |  |
|                                                                        | 5% WB       | MCDBR  | 5.1   | 5.9   | 6.7   | 7.1   | 8.6   | 8.2   | 8.8   | 8.6   | 7.2   | 8.0   | 6.5   | 5.9   |  |  |
|                                                                        |             | MCWBR  | 4.7   | 5.8   | 6.8   | 7.2   | 8.6   | 8.3   | 8.8   | 8.6   | 7.2   | 7.9   | 6.2   | 5.4   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.258 | 0.242 | 0.202 | 0.124 | N/A   | N/A   | N/A   | 0.124 | 0.191 | 0.216 | 0.235 | 0.254 |  |  |
|                                                                        | taud        |        | 2.332 | 2.339 | 2.262 | 1.913 | N/A   | N/A   | N/A   | 1.881 | 2.180 | 2.246 | 2.289 | 2.315 |  |  |
|                                                                        | Ebn at Noon |        | 954   | 892   | 774   | 312   | 0     | 0     | 0     | 280   | 768   | 920   | 979   | 979   |  |  |
|                                                                        | Edh at Noon |        | 95    | 76    | 51    | 11    | 0     | 0     | 0     | 11    | 53    | 82    | 98    | 102   |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 6.90  | 4.28  | 1.71  | 0.22  | 0.00  | 0.00  | 0.00  | 0.06  | 1.00  | 3.50  | 6.67  | 8.04  |  |  |
|                                                                        | RadStd      |        | 0.37  | 0.20  | 0.08  | 0.02  | 0.00  | 0.00  | 0.00  | 0.00  | 0.06  | 0.16  | 0.20  | 0.31  |  |  |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (0 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page